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Enhancing Well Integrity in the Middle East: Utilization of Novel Elastomeric Swelling Sleeve for Sustained Annular Pressure Prevention

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Abstract

Sustained Annular Pressure (SAP), a surface pressure buildup across one or multiple casing annuli as a result of fluid migration through permeating leaking barriers, originates from various factors including insufficient cement quality and progressive cement degradation over time. If not effectively managed, SAP can lead to critical well failures such as casing damage, thereby increasing the risk of well integrity failure and associated environmental impact.

This issue presents a significant concern for the global oil and gas industry, as approximately 50% of wells over 15 years old exhibit at least one casing annulus with SAP buildup worldwide (Yousuf et al., 2021). The situation is therefore particularly critical in the Middle East, where more than 50% of wells exceed 25 years of age. Consequently, oilfield operators across the Middle East face ongoing difficulties managing SAP in aging assets while implementing preventative measures for new wells. These circumstances highlight the urgent need for comprehensive mitigation strategies to ensure operational safety and extend the lifespan of well assets.

A novel and unique field-proven elastomeric swelling sleeve has been developed for the purpose of SAP prevention by sealing casing-to-cement micro annuli immediately upon exposure to fluid migration. Distinctly different from a swellable packer, the elastomeric swelling sleeve is a thin elastic seal designed for onsite slip-on installation onto oilfield tubulars. This high-value solution is capable of expanding and forming a reliable casing-to-cement seal when micro annular flow paths develop over time during the life of a well; it does not require additional downhole components, moving parts, or operational actuation for its activation. Developed in partnership with a major E&P company, the technology has been successfully field proven in over 500 global installations, including a 16-well shale-gas campaign.

The elastomeric swelling sleeve provides enhanced value by effectively preventing gas migration across critical cement barriers, thereby eliminating the need for remedial cementing throughout the well's lifecycle. This capability leads to substantial capital and operational expenditure savings by streamlining decommissioning operations, particularly due to the absence of SAP buildup at the time of plug & abandonment.

This publication offers a detailed description of the technology, highlighting its onsite installation process and examining its exceptional potential to mitigate SAP buildup occurrences across the Middle East region.

Additionally, it discusses the development and qualification of an innovative CO₂-swelling version of the technology, featuring the world's first purpose-built CO₂-swelling elastomer, specifically engineered to optimize well integrity in underground CO₂ storage wells and oilfield applications subjected to bottom-hole temperatures up to 200°C (392°F).

Introduction

Sustained Annular Pressure (SAP) represents a significant global environmental concern as it directly contributes to greenhouse gas emissions. SAP is also a clear indicator of well integrity issues as it signals the failure of one or more barriers within a well. Factors contributing to SAP emergence in oil & gas wells include inadequate cement placement, chemical degradation of cement, failure of downhole sealing equipment, and the effects of cyclic thermal expansion and pressure fluctuations, which place significant mechanical stress on cemented casing over time. A common denominator among these factors is that they all compromise casing-to-cement bond integrity, ultimately leading to the emergence of a micro-annular gap between the casing and the cement sheath. This micro-annulus can act as a conduit for gas migration, thereby increasing the likelihood of sustained annular pressure and subsequent SAP build up on surface during the life of the well.

It is important to note that SAP cannot be entirely and reliably prevented through conventional technologies. Even in cases where wells are constructed to the highest standards and cement is carefully planned and effectively placed, the development of micro-annuli remains a likely possibility as the well matures due to the long-term effects of mechanical, thermal, and chemical stresses acting on the cement sheath during the life of the well. Restoring well integrity after SAP emergence is often costly and time-consuming as remedial cementing in aged wells is more complex and risk prone in comparison to primary cementing, requiring extended rig time and increasing operational expenditure.

In the Middle East, while it is difficult to provide an accurate percentage estimate of oil & gas wells affected by SAP in the region, available studies suggest that a significant portion of older wells are critically impacted considering that more than 70% of the approximately 800 offshore platforms and their associated well stock in the Middle East are over 25 years old, placing them at elevated risk for well integrity issues such as SAP (Feather et al., 2018).

A new effective approach developed for preventing SAP build up involves the use of an elastomeric swelling sleeve, which was specifically engineered to establishing a permanent cement-to-casing bond at key locations along the wellbore. This field-proven technology prevents well integrity challenges and optimizes cement performance during the life of the well. This publication has been written to showcase the technology, provide an overview of its onsite installation procedure and track record, and discuss its high-value potential for well integrity optimization across the Middle East.

Technology Description

Designed specifically to prevent micro-annular fluid migration and SAP build up on surface, the elastomeric swelling sleeve is a one-of-a-kind downhole sealing device engineered from a highly elastic swelling elastomer. It activates upon contact with migrating fluids within the micro annulus, expanding and effectively sealing the gap between casing and cement sheath. With an average cross-sectional thickness of 1 mm, the elastomeric swelling sleeve features a slim cylindrical profile that enables it to be externally slipped over oilfield tubulars. Its compound formulation allows for both sufficient outward radial expansion to accommodate installation and adequate inward contraction to ensure a secure fit for save conveyance to setting depth without the need of adhesive chemicals. The primary objective of the technology is to ensure a durable cement-to-casing bond at strategically critical locations along the wellbore.

Unlike conventional swellable packers, the elastomeric swelling sleeve does not incorporate running guides or a metallic slip-on carrier sleeve. Due to its slim cross-sectional profile, its design is intended

exclusively for integration within the cement column, while ensuring an unobstructed annular flow path during cementing operations, which can help manage equivalent cement density (ECD) and reduce pressure-related risks during cementing. With a total expansion ratio capacity of 2:1, the elastomeric swelling sleeve is engineered to swell partially during deployment, thereby preserving expansion capacity for later activation. If a casing-to-cement micro-annulus forms and the elastomer is exposed to fluid migration, the sleeve will swell immediately to seal the micro-annular flow pathway, maintaining long-term zonal isolation during the life of a well. This expansion capacity is sufficient for cement assurance applications, as casing-to-cement micro-annuli typically ranges around ~1 mm in width. The following illustration offers an overview of the elastomeric swelling sleeve design:

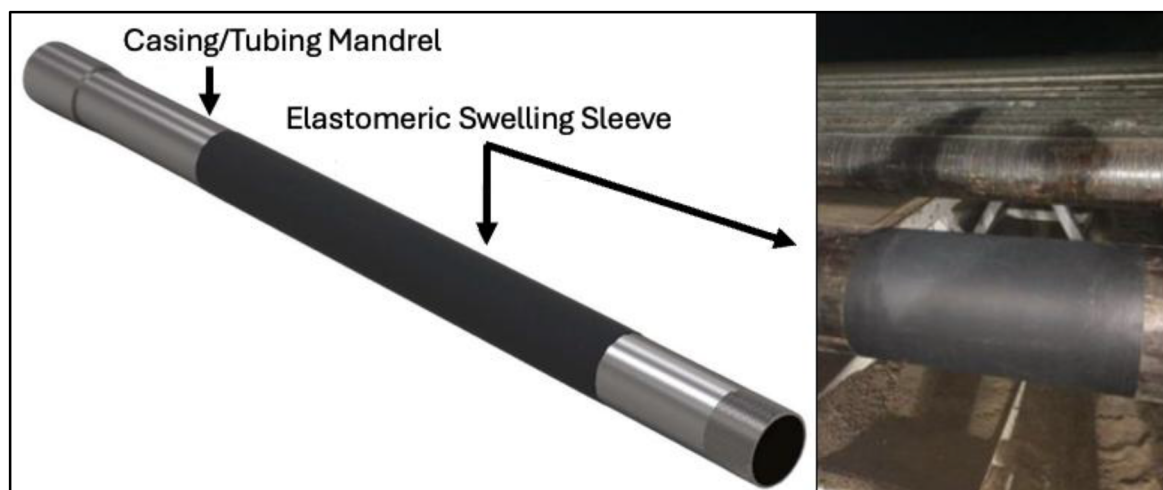


Figure 1—Elastomeric Swelling Sleeve.

Chemically compatible with water and oil-based fluids, the elastomeric swelling sleeve is rated for continuous service at temperatures up to 150°C (302°F). It is compatible with all standard oilfield tubing and casing sizes and has demonstrated successful deployment across a range of casing applications, including surface, intermediate, and production liner strings.

The elastomeric swelling sleeve installation is carried out without supplementary downhole components, mechanical workstring actuation, or downhole moving parts, enabling passive activation through direct fluid exposure. By effectively sealing gas migration flow pathways while incurring minimal operational time and associated costs, the elastomeric swelling sleeve delivers substantial value, particularly through the simplification of plug and abandonment (P&A) operations during well decommissioning. Its ability to maintain long term well integrity eliminates the need for remedial cementing techniques such as perforate wash & cement (PwC) or section milling abandonment operations, which typically require the utilization of a rig. The following figure illustrates possible installation points along the wellbore for both casing and tubing strings.

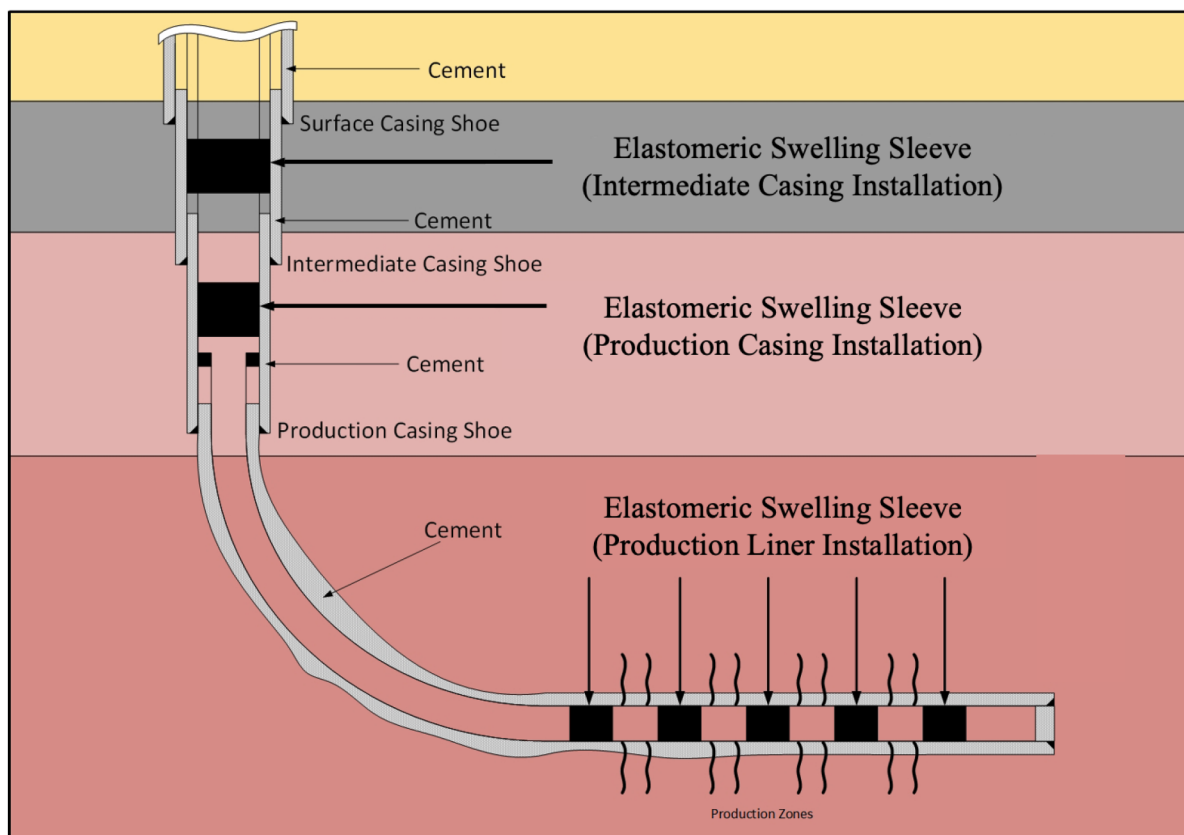


Figure 2—Elastomeric Swelling Sleeve Installation onto Multiple Tubular Strings.

Product Qualification

The elastomeric swelling sleeve was developed through a collaborative effort between a major global oil & gas operating company and a Netherlands-based completion equipment supplier, enabling accelerated laboratory and field product qualification ahead of its 2023 commercial product launch. Product qualification testing was conducted to validate key performance attributes during the developmental phase, which included:

- **Material Longevity Testing:** Carried out at a reputable Netherlands-based independent research organization. Results confirmed long lasting material properties exceeding 80-year functionality.
- **Swell Rate Test:** Elastomer samples were tested in KCl brine systems at different salinity concentrations and temperatures in order to study the effects of these parameters on swell rate at downhole conditions. Results were comparable with industry standard water-reactive swellable packer technology. More specifically, swell rate was confirmed to be directly proportional to temperature and inversely proportional to salinity concentration.
- **Micro Annular Sealing Test:** Two vertical test fixtures configured with differential pressure testing capability were both filled with Portland Class G cement. One fixture had an elastomeric swelling sleeve and the other had cement only (no elastomeric swelling sleeve). The objective of this test was to create a micro annular gap across the cement column within both test fixtures and evaluate the elastomeric swelling sleeve seal within the micro annular gap. The test confirmed that the cement-only fixture developed a micro-annulus void allowing crossflow with a flow rate of 700 ml/min and could not hold differential pressure. The fixture with an elastomeric swelling sleeve effectively sealed the void and withstood 20 bar (290 psi) differential pressure. The following provides an overview of the test setup and differential pressure testing results.

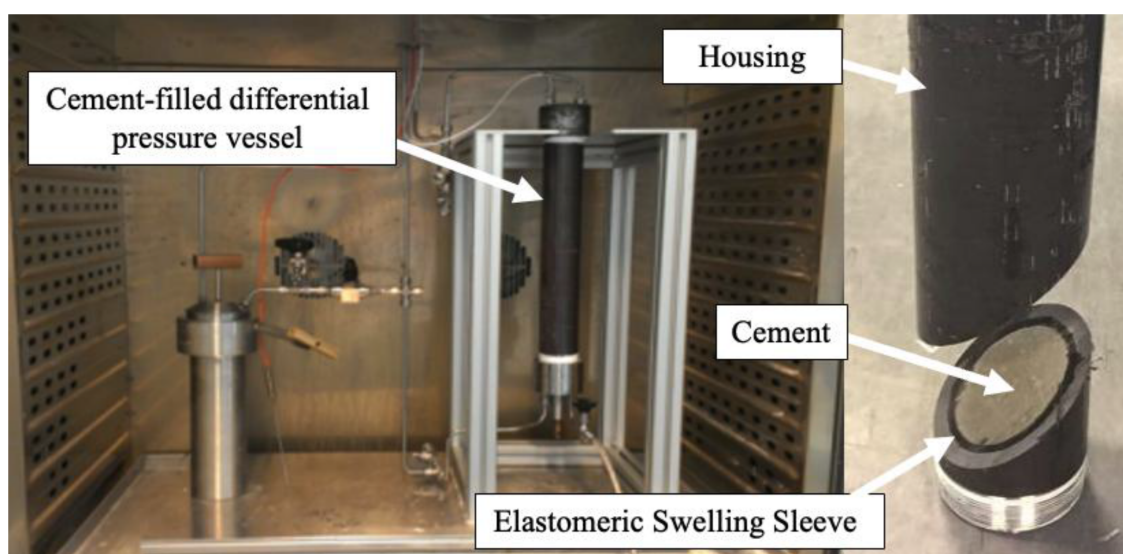


Figure 3—Micro annular seal pilot lab test setup.

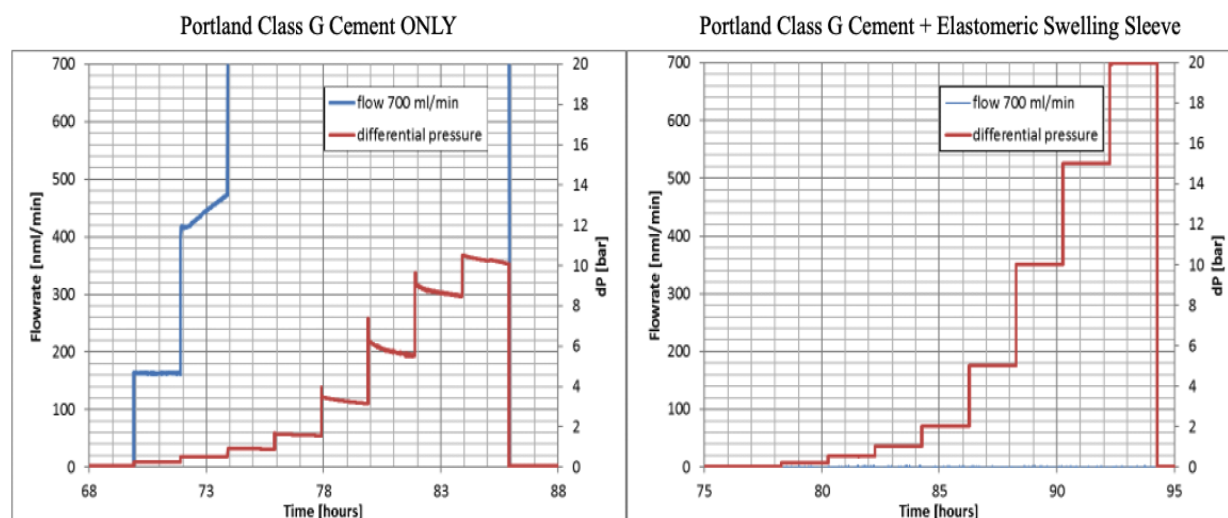


Figure 4—Micro annular seal pilot lab test results (with & without the elastomeric swelling sleeve).

Onsite Installation Process

The elastomeric swelling sleeve installation is recommended to take place on location in the field. The installation process requires the following items:

- A sleeve stretcher: A cylindric scabbard configured with a handle and an air pump connection port located in the center of the stretcher. The purpose of the stretcher is to obtain sufficient outward radial expansion to enable slip-on installation.
- Removable clamps: Required to enable stretching the elastomeric swelling sleeve ends across the circumference of the stretcher. These clamps hold the folded sleeve securely in place, preventing any shifting or sliding that could occur when the sleeve ends are wrapped around the stretcher.
- Dual-sided adhesive tape: Required to keep the elastomeric swelling sleeve fixed onto the stretcher ends during slip-on installation.
- Manual air pump: Required to vacuum out air excess between the elastomeric swelling sleeve and the stretcher, allowing for a smooth internal circumferential profile which enables slip-on installation onto casing or tubing.

During the first step, the elastomeric swelling sleeve is inserted inside the stretcher and folded across both ends with the assistance of dual sided adhesive tape and one clamp. The following images provide a graphic overview of the initial installation step.

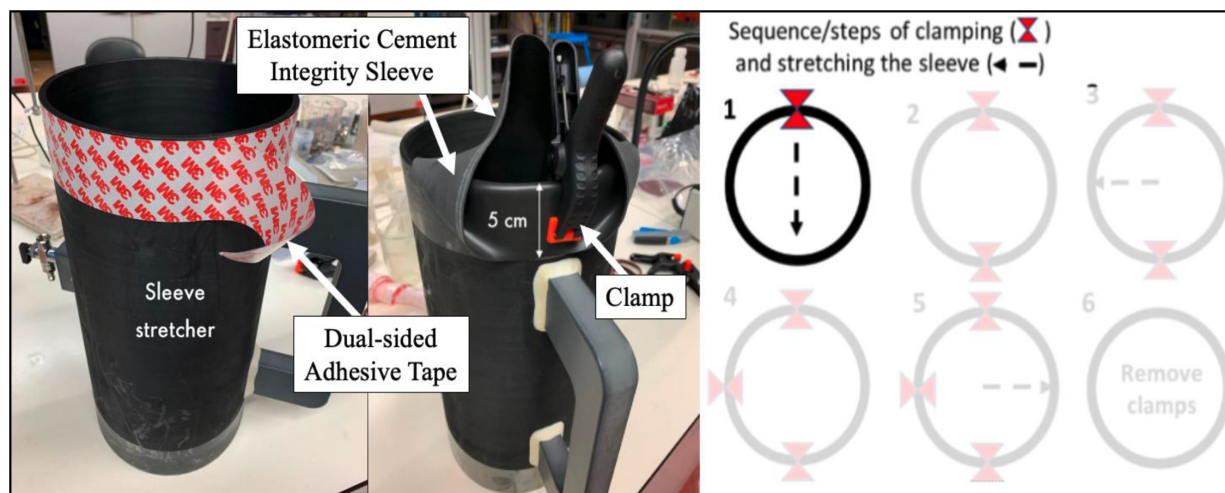


Figure 5—Initial installation step.

The elastomeric swelling sleeve is intentionally longer than the stretcher itself. This extra length allows for approximately 5 cm of the sleeve to be folded over each end of the stretcher. Folding the sleeve over both ends is essential for creating an airtight seal, enabling the hand air pump to effectively vacuum out excess air and expand the sleeve evenly around the stretcher's internal circumference, creating a smooth internal profile to enable slip-on installation. The use of removable clamps is crucial during this process. Attempting installation without clamps may result in misalignment or accidental unfolding at either end. The following images visually demonstrate this installation step.

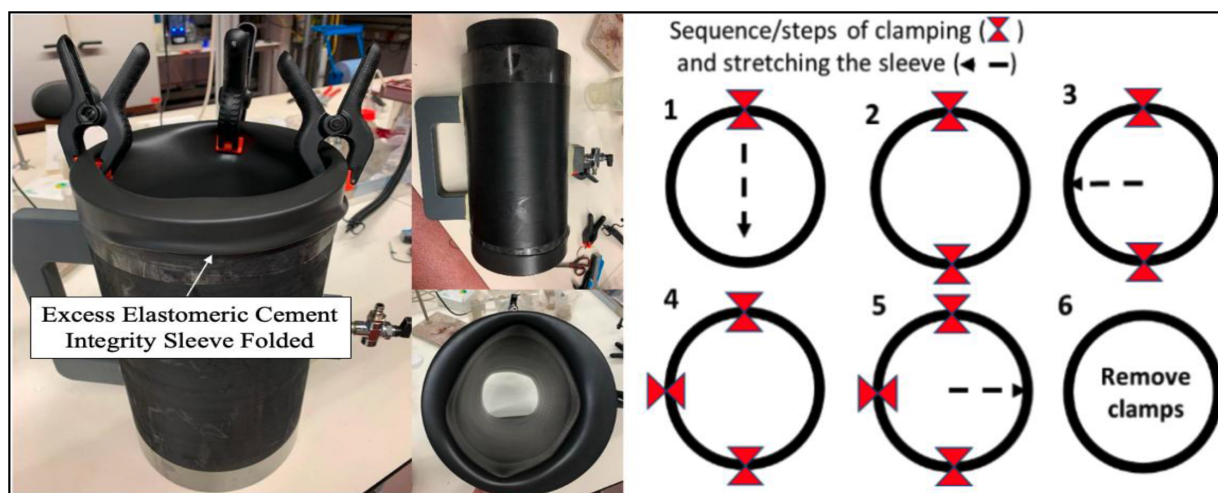


Figure 6—Installation inside stretcher (first end fold).

After inserting and folding the elastomeric swelling sleeve over the top end of the stretcher, the stretcher is inverted in order to carry out identical folding on the sleeves opposite end. When both ends have been folded in this manner, the elastomeric swelling sleeve should resemble the setup depicted in the following image. At this stage, the elastomeric swelling sleeve is prepared for vacuum air extraction inside the stretcher.

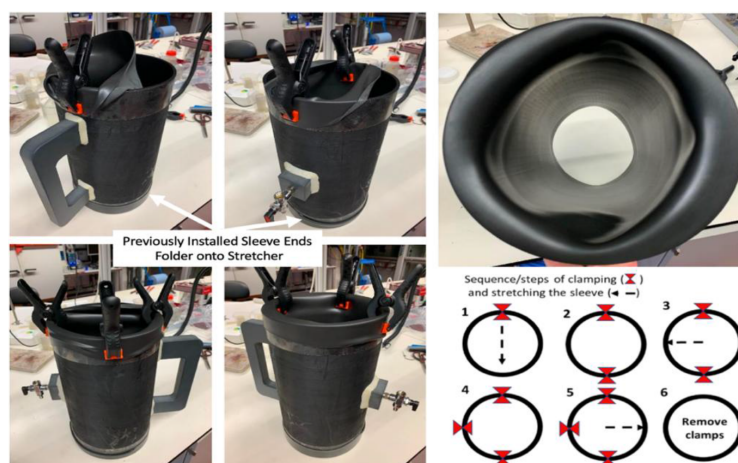


Figure 7—Installation inside stretcher (second end fold).

In the next step, the manual air pump is connected to the stretcher's valve port, while ensuring that the valve is in the "open" position, as illustrated in the following image.

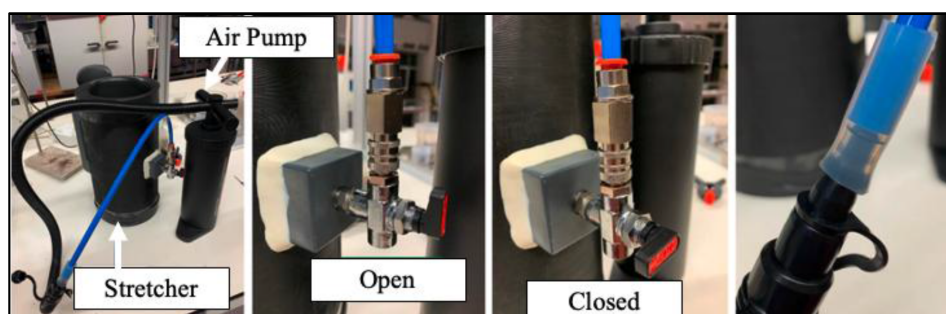


Figure 8—Stretcher connected to hand air pump (vacuum suction).

After the elastomeric swelling sleeve has been fitted inside the stretcher and the manual air pump is attached to the stretcher valve, the air trapped between the sleeve and the stretcher's inner walls is vacuumed out. This process causes the sleeve to expand evenly throughout the interior. Minor air pockets may be observed at either end of the stretcher, which do not prevent installation. Once this step is complete, the stretcher's valve is closed, and the manual pump is disconnected from the stretcher. The following image illustrates the internal appearance of the elastomeric swelling sleeve both before and after vacuum suction is applied.



Figure 9—Internal profile before and after excess air removal.

With the elastomeric swelling sleeve fully expanded inside the stretcher, the entire assembly is then slid externally over the casing or tubing mandrel to the specified location according to pipe tally requirements.

Once the correct position is reached, open the stretcher valve to release the vacuum pressure. This action allows the elastomeric swelling sleeve to relax and form a secure fit around the base pipe. At this stage, the stretcher can be removed, as illustrated in the following image.

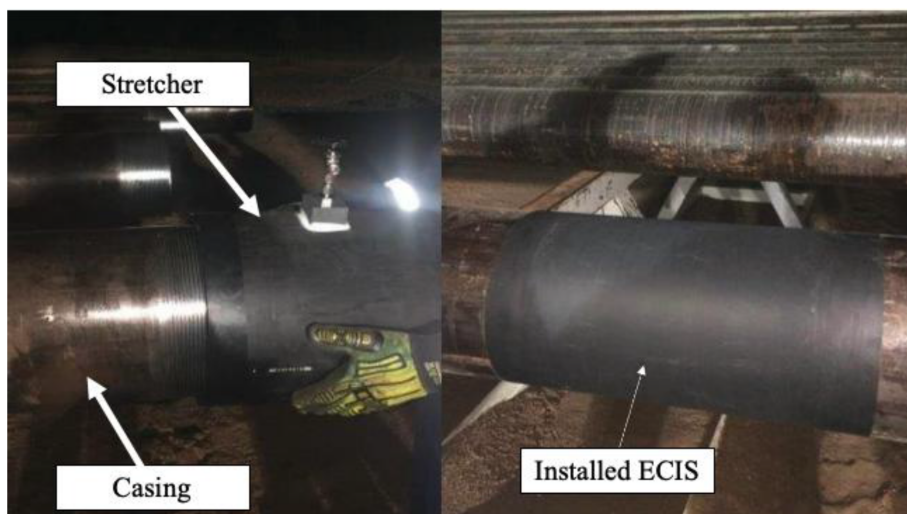


Figure 10—Stretcher removed & elastomeric swelling sleeve installed.

It is recommended to use a centralizer secured with a stop collar near the elastomeric swelling sleeve to enhance overall cement job quality and provide stand-off protection during conveyance to setting depth. Additionally, considering the reduced cross-sectional thickness of the elastomeric swelling sleeve, installations located within 5 feet from a casing coupling would have sufficient stand-off protection preventing the elastomeric swelling sleeve from rotating or rolling over, particularly when there is a risk of dragging through deviated sections or encountering tight clearance restrictions along the well path. The image below illustrates the recommended placement of the centralizer in relation to the elastomeric swelling sleeve.

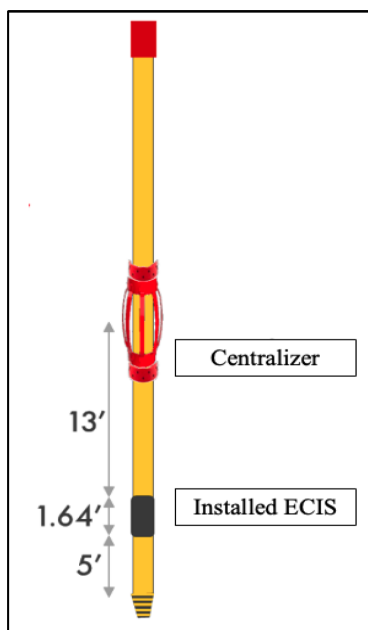


Figure 11—Recommended centralization and position with respect to lower coupling.

Track Record

The elastomeric swelling sleeve has a strong demonstrated global track record exceeding 500 successful installations with no reported operational issues to date. One of the most notable demonstrations of its optimal performance occurred during a 16-well case study carried out in the Far East. In this project, every well was drilled in a dedicated shale gas field, with each well designed as a shale gas producer utilizing a monobore completion featuring 4.5" production casing run continuously from total depth to surface. The technology was introduced specifically to mitigate recurring SAP problems which had affected previous hydraulic fracturing well campaigns operations in the same field with similar well configurations. By incorporating the elastomeric swelling sleeve into the well design, the operator achieved an 80% reduction in SAP incidents in comparison to previous identical well campaigns where the technology had not been utilized.

This improvement not only enhanced well integrity across the board but also resulted in substantial cost savings by eliminating remedial cementing and workover operations, estimated at more than \$2 million for the campaign. Documented in SPE paper IPTC-24028, this case study highlights the elastomeric swelling sleeve's ability to maintain cement sheath integrity and prevent SAP even under the cyclic pressure and temperature variations commonly encountered in multi-stage hydraulic fracturing operations, which was verified several months after original hydraulic fracturing operations. These results establish the elastomeric swelling sleeve as a proven solution for enhancing well performance and integrity across a range of reservoir conditions.

First Installation in the Middle East Region

The elastomeric swelling sleeve was successfully installed for the first time in the Middle East in April 2025, following its regional introduction at the 2024 International Petroleum Technology Conference in Saudi Arabia. The installation took place in a high-pressure, high-temperature extended-reach well located within an unconventional gas reservoir in a tight onshore gas field, where it was mounted onto 7-5/8" intermediate casing and cemented in place at an approximate setting depth of 750 meters (2,460 feet).

This deployment was particularly significant as it took place in one of the deepest tight gas plays globally, where extreme downhole conditions demand advanced engineering solutions. In this application, downhole conditions include exposure to corrosive fluid environments and frequent thermal and pressure cycling associated with shut-in, stimulation, and production operations. These factors threaten long-term cement sealing effectiveness as they increase the risk of casing-to-cement micro-annuli, fluid migration, and subsequent SAP build up on surface.

In view of the aforementioned challenges, a leading national operating company opted to utilize the elastomeric swelling sleeve in order to mitigate the emergence of SAP as these harsh subsurface conditions collectively impose stringent requirements on cementing performance and long-term well integrity, making reliable zonal isolation across cement barriers essential to operational success. The introduction of the elastomeric swelling sleeve in the region addresses these challenges by enhancing cement barrier sealing capacity in demanding environments, marking a significant advancement in well integrity management within the Middle East. The following image shows the first installation of the technology in the Middle East.



Figure 12—First Middle East Installation

It is important to note that occurrences of sustained annular pressure (SAP) typically take time to develop. Therefore, the well must be allowed to age in order to quantify long-term performance. Continuous monitoring will be conducted following fracturing operations, with particular attention to SAP emergence over time. In order to ensure an effective evaluation, results from this well will be eventually compared against nearby wells with similar designs and operating conditions within the same field. Once sufficient time has elapsed and further observations are gathered, a follow-up will be published to provide the industry with an informed update. The aim of publishing this paper at this stage is to raise awareness across the region about the strong track record of this field-proven technology and to share news of its first installation in the Middle East.

Feasibility and Benefits for Carbon Capture and Storage Applications

Carbon Capture and Storage (CCS) operations require robust well integrity planning and maintenance to ensure the safe and effective downhole containment of CO₂, thereby preventing environmental hazards. A significant challenge in CCS wells is the degradation of cement due to chemical attack by carbonic acid, which forms through multiple reactions between CO₂ and water under supercritical conditions. This degradation increases the risk of SAP emergence on surface and potential leakage from CCS wells into the atmosphere.

To address this issue, a novel CO₂-swelling version of the technology has been developed specifically for CCS environments. Rated for temperatures up to 200°C (392°F), this technology features the world's first purpose-built CO₂-swelling elastomer which reliably expands in the presence of both gaseous and aqueous CO₂, establishing an effective barrier even under aggressive chemical conditions. The adoption of this technology in CCS applications has the potential to significantly enhance well integrity and mitigate the risk of CO₂ leakages in long-term carbon storage operations.

Given its higher temperature rating, the CO₂-swelling version of the technology is also well suited for oil and gas applications which exceed the 150°C (302°F) temperature limit of the conventional elastomeric swelling sleeve discussed in this paper. Its combination of thermal stability and chemical resistance makes it a strong candidate for both CCS and demanding high temperature oil and gas applications.

Conclusion

Many oilfields across the Middle East increasingly encounter well integrity challenges due to aging infrastructure, complex well designs, and harsh reservoir conditions. Implementing the utilization of the elastomeric swelling sleeve across the region provides a proactive solution by offering reliable zonal isolation and SAP prevention. A key advantage of this technology is its preventive nature as it is implemented at the well planning phase, enabling operators to address well integrity concerns before they arise.

The elastomeric swelling sleeve offers an unmatched value proposition withing the oil and gas industry as its relatively low in cost in comparison to traditional remedial intervention operations while considering it has been field-proven to eliminate SAP occurrences by up to 80%. The technology serves as a cost-effective insurance policy for safeguarding well integrity from the planning phase. Given these compelling benefits, the widespread adoption of this novel technology is strongly recommended for operators throughout the Middle East seeking to enhance well integrity, reduce operational risks, and ensure long-term asset performance.

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